

# Differentiation test · Algebra

Grade 11

40 minutes 100 marks 15 questions + bonus No calculator

- 3–4 The limit of a function — the 0/0 form
- 7–9 The rules of differentiation
- 10–12 The derivative of a composite function
- 17–18 The equations of the tangent and the normal

Algebra 11, §1.2 · P1 · 7.1  
 Algebra 11, §1.4 · P1 · 7.2–7.4  
 Algebra 11, §1.5 · P1 · 7.5  
 Algebra 11, §1.6 · P1 · 7.6

- Answer on separate paper, and show your working — method carries marks.
- Simplify every derivative. Leave no negative or fractional index in a final answer.

## What the paper weighs

| TOPIC                                                       | MARKS | SHARE |
|-------------------------------------------------------------|-------|-------|
| Lessons 3–4 · The limit of a function                       | 8     | 8%    |
| Lessons 7–9 · The rules of differentiation                  | 46    | 46%   |
| Lessons 10–12 · The derivative of a composite function      | 36    | 36%   |
| Lessons 17–18 · The equations of the tangent and the normal | 10    | 10%   |

The rules and the chain rule carry 82 of the 100 marks between them, which is what this paper is for. The two 0/0 limits and the tangent are there so that a derivative is used for something and not only performed. Nothing on the paper needs lessons 1–2, 5–6, 15–16, 19–22 or 23–25.

## A Section A · Five marks each

EASY 40 marks

Differentiate and simplify. One line of working is enough.

1  $y = 4x^3 - 5x + 7$  [5]

ANSWER

$$\frac{dy}{dx} = 12x^2 - 5$$

WORKING

Power rule term by term; the derivative of the constant 7 is zero.

2  $y = \frac{3}{x^2}$  [5]

ANSWER

$$\frac{dy}{dx} = -\frac{6}{x^3}$$

WORKING

Rewrite as  $y = 3x^{-2}$ , so  $\frac{dy}{dx} = -6x^{-3} = -\frac{6}{x^3}$ .

The rewrite is the whole question. A student who cannot turn  $\frac{3}{x^2}$  into a power cannot start.

3  $y = 5\sqrt{x}$  [5]

ANSWER

$$\frac{dy}{dx} = \frac{5}{2\sqrt{x}}$$

WORKING

Rewrite as  $y = 5x^{1/2}$ , so  $\frac{dy}{dx} = \frac{5}{2}x^{-1/2} = \frac{5}{2\sqrt{x}}$ .

4  $y = \frac{x^3 - 2x}{x}$  [5]

ANSWER

$$\frac{dy}{dx} = 2x$$

WORKING

Divide first:  $y = x^2 - 2$  for  $x \neq 0$ , so  $\frac{dy}{dx} = 2x$ .

The quotient rule works but wastes two minutes. Looking at the expression before reaching for a rule is the point.

5  $y = (3x + 1)^5$  [5]

ANSWER

$$\frac{dy}{dx} = 15(3x + 1)^4$$

WORKING

Chain rule:  $5(3x + 1)^4$  times the derivative of the inside, which is 3.

6  $y = (x^2 - 4)^3$  [5]

ANSWER

$$\frac{dy}{dx} = 6x(x^2 - 4)^2$$

WORKING

Chain rule:  $3(x^2 - 4)^2$  times the derivative of the inside, which is  $2x$ .

7  $y = \frac{1}{2x - 5}$  [5]

ANSWER

$$\frac{dy}{dx} = -\frac{2}{(2x - 5)^2}$$

WORKING

Rewrite as  $y = (2x - 5)^{-1}$ , so  $\frac{dy}{dx} = -(2x - 5)^{-2} \cdot 2 = -\frac{2}{(2x - 5)^2}$ .

8  $y = \sqrt{4x + 1}$  [5]

ANSWER

$$\frac{dy}{dx} = \frac{2}{\sqrt{4x + 1}}$$

WORKING

Rewrite as  $y = (4x + 1)^{1/2}$ , so  $\frac{dy}{dx} = \frac{1}{2}(4x + 1)^{-1/2} \cdot 4 = \frac{2}{\sqrt{4x + 1}}$ .

## B Section B · Eight marks each

MEDIUM

40 marks

Method marks are available even when the final answer is wrong.

9 Differentiate  $y = \frac{2x - 1}{x^2 + 3}$  and find the gradient of the curve at  $x = 1$ . [8]

ANSWER

$$\frac{dy}{dx} = \frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}; \text{ gradient} = \frac{3}{8}$$

WORKING

Quotient rule with  $u = 2x - 1$ ,  $v = x^2 + 3$ , so  $u' = 2$  and  $v' = 2x$ :

$$\frac{dy}{dx} = \frac{2(x^2 + 3) - (2x - 1)(2x)}{(x^2 + 3)^2} = \frac{2x^2 + 6 - 4x^2 + 2x}{(x^2 + 3)^2} = \frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}.$$

At  $x = 1$ :  $\frac{-2 + 2 + 6}{4^2} = \frac{6}{16} = \frac{3}{8}$ .

The expansion  $-(2x - 1)(2x) = -4x^2 + 2x$  is where the sign is usually lost.

10  $y = x^3 - 3x^2 + 4$ . Find the values of  $x$  at which the gradient of the curve is 9. [8]

ANSWER

$$x = 3 \text{ and } x = -1$$

WORKING

$$\frac{dy}{dx} = 3x^2 - 6x. \text{ Set it equal to 9: } 3x^2 - 6x = 9, \text{ so } 3x^2 - 6x - 9 = 0 \text{ and } x^2 - 2x - 3 = 0.$$

That factorises as  $(x - 3)(x + 1) = 0$ , giving  $x = 3$  and  $x = -1$ .

Two answers, not one. Dividing by 3 before factorising saves the arithmetic.

11  $y = \sqrt{x^2 + 9}$ . Find  $\frac{dy}{dx}$  and its value at  $x = 4$ . [8]

ANSWER

$$\frac{dy}{dx} = \frac{x}{\sqrt{x^2 + 9}}; \text{ at } x = 4 \text{ it is } \frac{4}{5}$$

WORKING

Write  $y = (x^2 + 9)^{1/2}$ . Chain rule:  $\frac{dy}{dx} = \frac{1}{2}(x^2 + 9)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2 + 9}}$ .

At  $x = 4$ :  $\sqrt{16 + 9} = 5$ , so the value is  $\frac{4}{5}$ .

The factor  $2x$  from the inside function is the mark most often dropped.

12 Differentiate  $y = \frac{5}{(1 - 2x)^3}$ . [8]

ANSWER

$$\frac{dy}{dx} = \frac{30}{(1 - 2x)^4}$$

WORKING

Rewrite as  $y = 5(1 - 2x)^{-3}$ . Chain rule:

$$\frac{dy}{dx} = 5 \cdot (-3)(1 - 2x)^{-4} \cdot (-2) = 30(1 - 2x)^{-4} = \frac{30}{(1 - 2x)^4}.$$

Two minus signs, and they cancel: the  $-3$  from the power and the  $-2$  from the inside. An answer of  $-\frac{30}{(1 - 2x)^4}$  means only one of them was used.

13 Evaluate:

(a)  $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$  [4]

(b)  $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$  [4]

ANSWER

(a) 6    (b)  $\frac{1}{4}$

WORKING

(a) Substitution gives  $\frac{0}{0}$ , so factorise:  $\frac{(x-3)(x+3)}{x-3} = x+3$  for  $x \neq 3$ , which tends to 6.

(b)  $\frac{0}{0}$  again, with a root — multiply by the conjugate  $\sqrt{x+4} + 2$ :

$$\frac{(x+4) - 4}{x(\sqrt{x+4} + 2)} = \frac{x}{x(\sqrt{x+4} + 2)} = \frac{1}{\sqrt{x+4} + 2}, \text{ for } x \neq 0, \text{ which tends to } \frac{1}{2+2} = \frac{1}{4}.$$

The two techniques for  $\frac{0}{0}$ : factorise and cancel, or multiply by the conjugate.  $\frac{0}{0}$  written down as an answer scores nothing — it is the signal that algebra is needed, not a value.

## C Section C • Ten marks each

Full method required.

HARD

20 marks

14  $y = x^2\sqrt{2x-1}$ .

(a) Show that  $\frac{dy}{dx} = \frac{x(5x-2)}{\sqrt{2x-1}}$ . [6]

(b) Hence find the gradient of the curve at  $x = 5$ . [4]

ANSWER

(a) shown below (b)  $\frac{115}{3}$

WORKING

(a) Product rule with  $u = x^2$  and  $v = (2x-1)^{1/2}$ . By the chain rule  $v' = \frac{1}{2}(2x-1)^{-1/2} \cdot 2 = \frac{1}{\sqrt{2x-1}}$ , so

$$\frac{dy}{dx} = 2x\sqrt{2x-1} + \frac{x^2}{\sqrt{2x-1}}.$$

Over the common denominator  $\sqrt{2x-1}$ :  $\frac{2x(2x-1) + x^2}{\sqrt{2x-1}} = \frac{4x^2 - 2x + x^2}{\sqrt{2x-1}} = \frac{5x^2 - 2x}{\sqrt{2x-1}} = \frac{x(5x-2)}{\sqrt{2x-1}}.$

(b) At  $x = 5$ :  $\sqrt{10-1} = 3$  and  $x(5x-2) = 5 \cdot 23 = 115$ , so the gradient is  $\frac{115}{3}$ .

The product rule and the chain rule in one question, and then the algebra that puts the two terms over one denominator — which is where the “show that” is won or lost.

15 The curve  $y = ax^2 + bx$  passes through  $(2, 2)$  and has gradient 5 at  $x = 2$ .

(a) Find  $a$  and  $b$ . [7]

(b) Hence find the equation of the tangent to the curve at  $(2, 2)$ . [3]

ANSWER

(a)  $a = 2, b = -3$  (b)  $y = 5x - 8$

WORKING

(a) Two facts give two equations.

The curve passes through  $(2, 2)$ :  $4a + 2b = 2$ , so  $2a + b = 1$ .

The gradient is  $\frac{dy}{dx} = 2ax + b$ , and at  $x = 2$  it is 5:  $4a + b = 5$ .

Subtracting the first from the second gives  $2a = 4$ , so  $a = 2$ , and then  $b = 1 - 4 = -3$ . The curve is  $y = 2x^2 - 3x$ .

(b) The gradient at  $(2, 2)$  is 5, so  $y - 2 = 5(x - 2)$ , that is  $y = 5x - 8$ .

“Passes through” is a statement about  $y$ , “has gradient” is a statement about  $\frac{dy}{dx}$  — turning each into its own equation is the whole question. A check is worth the thirty seconds:  $y(2) = 8 - 6 = 2$  and  $y'(2) = 8 - 3 = 5$ .

### Bonus

★ Attempt this only when the rest is finished. It can make up marks lost elsewhere, but your total is still recorded out of 100. BONUS +10 marks

16  $y = \frac{x+1}{\sqrt{x}}$  for  $x > 0$ .

(a) Show that  $\frac{dy}{dx} = \frac{x-1}{2x\sqrt{x}}$ . [7]

(b) Hence find the gradient of the curve at  $x = 4$ . [3]

ANSWER

(a) shown below    (b)  $\frac{3}{16}$

WORKING

(a) Split the fraction before differentiating:

$$y = \frac{x}{\sqrt{x}} + \frac{1}{\sqrt{x}} = x^{1/2} + x^{-1/2}.$$

$$\text{Then } \frac{dy}{dx} = \frac{1}{2}x^{-1/2} - \frac{1}{2}x^{-3/2} = \frac{1}{2}x^{-3/2}(x-1) = \frac{x-1}{2x\sqrt{x}}, \text{ since } x^{3/2} = x\sqrt{x}.$$

(b) At  $x = 4$ :  $x-1 = 3$  and  $2x\sqrt{x} = 2 \cdot 4 \cdot 2 = 16$ , so the gradient is  $\frac{3}{16}$ .

The quotient rule also works, but it takes twice as long and the simplification at the end is harder. Splitting the fraction first is the skill being tested.